

NE523 Homework 2

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Question 1 Fluxes & Currents

Angular flux $\psi(\hat{\Omega})$ is the intensity of neutrons in the direction $\hat{\Omega}$. In this case, since the intensity is I when in direction $\hat{i}$ (the unit vector in the x -direction), and zero otherwise. So,

$$\psi(\hat{\Omega}) = I\delta(\hat{i} \cdot \hat{\Omega} - 1), \quad (1)$$

since $\hat{i} \cdot \hat{\Omega} = 1$ when $\hat{\Omega} = \hat{i}$. Then, using this angular flux,

$$\varphi = \int_{4\pi} d\Omega I\delta(\hat{i} \cdot \hat{\Omega} - 1) = I. \quad (2)$$

The current vector is

$$\mathbf{J} = \int_{4\pi} d\Omega \hat{\Omega} \psi = \hat{i} I, \quad (3)$$

and the partial currents are

$$J_x^+ = \int_{\hat{\Omega} \cdot \hat{i} > 0} d\Omega \hat{i} \cdot \hat{\Omega} \psi = I \quad (4)$$

$$J_x^- = \int_{\hat{\Omega} \cdot \hat{i} < 0} d\Omega \hat{i} \cdot \hat{\Omega} \psi = 0 \quad (5)$$

Question 2 Slab Geometry Transport Solution

The transport equation for a purely absorbing medium with isotropic source is

$$\hat{\Omega} \cdot \nabla \psi(\mathbf{r}, \hat{\Omega}, E) + \sigma_t \psi = \frac{1}{4\pi} q(E). \quad (6)$$

Then, since in slab geometry $\frac{d\psi}{dy} = \frac{d\psi}{dz} = 0$, this becomes

$$\mu \frac{d}{dx} \psi(x, \hat{\Omega}, E) + \sigma_t \psi = \frac{1}{4\pi} q(E). \quad (7)$$

Since this is a separable ODE, the solution can be found by rearranging and integrating both sides:

$$-\mu \frac{d\psi}{dx} = \sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E) \quad (8)$$

$$\frac{1}{\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E)} d\mu = \frac{-1}{\mu} dx \quad (9)$$

$$\int \frac{1}{\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E)} d\mu = - \int \frac{1}{\mu} dx \quad (10)$$

$$\frac{\ln \left(\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E) \right)}{\sigma_t} = -\frac{x}{\mu} + c_0 \quad (11)$$

$$\sigma_t \psi(x, \hat{\Omega}, E) - \frac{1}{4\pi} q(E) = c_1 e^{-\frac{\sigma_t}{\mu} x} \quad (12)$$

$$\sigma_t \psi(x, \hat{\Omega}, E) = c_1 e^{-\frac{\sigma_t}{\mu} x} + \frac{1}{4\pi} q(E) \quad (13)$$

$$\psi(x, \hat{\Omega}, E) = c_2 e^{-\frac{\sigma_t}{\mu} x} + \frac{1}{4\pi \sigma_t} q(E). \quad (14)$$

The boundary conditions are different for different values of μ ; for $\mu > 0$, the constant $c_2 = 0$ since ψ must be finite for the left half of the domain. For $\mu < 0$, the incoming flux boundary condition $\psi(0, \mu < 0) = 0$ can be used, and the constant becomes $c_2 = -\frac{1}{4\pi \sigma_t} q(E)$. Then in the slab, the flux is

$$\psi(x, E) = \begin{cases} \frac{1}{4\pi \sigma_t} q(E), & \mu > 0 \\ \frac{1}{4\pi \sigma_t} q(E) \left(1 - e^{\frac{\sigma_t}{\mu} x} \right), & \mu < 0 \end{cases}, \quad -\infty \leq x < 0. \quad (15)$$

At the interface $x = 0$, since there is no incoming flux from the vacuum,

$$\psi(0, \Omega, E) = \begin{cases} \frac{1}{4\pi \sigma_t} q(E), & \mu \geq 0 \\ 0, & \mu < 0 \end{cases} \quad (16)$$

and the current vector is

$$\mathbf{J}(0, E) = \int_{4\pi} d\Omega \hat{\Omega} \psi = \hat{i} \frac{1}{2} q(E) \quad (17)$$

Question 3 Inner Products & Orthogonality

For the set of polynomials to be orthonormal on $x \in [-1, 1]$, they must satisfy the condition that $\langle P_i, P_j \rangle = \int_{-1}^1 P_i P_j dx = \delta_{i,j}$:

$$\begin{aligned}\langle P_0, P_0 \rangle &= \int_{-1}^1 \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} dx \\ &= \frac{1}{2} \int_{-1}^1 dx \\ \langle P_0, P_0 \rangle &= \boxed{1}\end{aligned}\tag{18}$$

$$\begin{aligned}\langle P_0, P_1 \rangle &= \int_{-1}^1 \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{\sqrt{2}} x dx \\ &= \frac{1}{2} \int_{-1}^1 x dx \\ &= \frac{1}{2} \left[\frac{x^2}{2} \right]_{-1}^1 \\ &= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{2} \right] \\ \langle P_0, P_1 \rangle &= \boxed{0}\end{aligned}\tag{19}$$

$$\begin{aligned}\langle P_0, P_2 \rangle &= \int_{-1}^1 \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{5}}{\sqrt{2}} \cdot \frac{1}{2} (3x^2 - 1) dx \\ &= \frac{\sqrt{5}}{4} \int_{-1}^1 (3x^2 - 1) dx \\ &= \frac{\sqrt{5}}{4} [x^3 - x]_{-1}^1 \\ &= \frac{\sqrt{5}}{4} [(1 - 1) - (-1 + 1)] \\ \langle P_0, P_2 \rangle &= \boxed{0}\end{aligned}\tag{20}$$

$$\begin{aligned}
\langle P_1, P_1 \rangle &= \frac{3}{2} \int_{-1}^1 x^2 dx \\
&= \frac{3}{2} \left[\frac{x^3}{3} \right]_{-1}^1 \\
&= \frac{3}{2} \left[\frac{1}{3} - \frac{-1}{3} \right]_{-1}^1 \\
\langle P_1, P_1 \rangle &= \boxed{1}
\end{aligned} \tag{21}$$

$$\begin{aligned}
\langle P_1, P_2 \rangle &= \sqrt{\frac{3}{2}} \sqrt{\frac{5}{2}} \int_{-1}^1 \frac{1}{2} x (3x^2 - 1) dx \\
&= \frac{\sqrt{15}}{2} \int_{-1}^1 \frac{3}{4} x^2 - \frac{1}{2} x dx \\
&= \frac{\sqrt{15}}{2} \left[\frac{3}{8} x^4 - \frac{1}{4} x^2 \right]_{-1}^1 \\
&= \frac{\sqrt{15}}{2} \left[\left(\frac{3}{8} - \frac{1}{4} \right) - \left(\frac{3}{8} - \frac{1}{4} \right) \right] \\
\langle P_1, P_2 \rangle &= \boxed{0}
\end{aligned} \tag{22}$$

$$\begin{aligned}
\langle P_2, P_2 \rangle &= \frac{5}{2} \int_{-1}^1 \frac{1}{2^2} (3x^2 - 1)^2 dx \\
&= \frac{5}{8} \int_{-1}^1 9x^4 - 6x^2 + 1 dx \\
&= \frac{5}{8} \left[\frac{9}{5} x^5 - 2x^3 + x \right]_{-1}^1 \\
&= \frac{5}{8} \left[\left(\frac{9}{5} - 2 + 1 \right) - \left(\frac{-9}{5} + 2 - 1 \right) \right] \\
\langle P_2, P_2 \rangle &= \boxed{1}.
\end{aligned} \tag{23}$$

This confirms the condition that $\langle P_i, P_j \rangle = \delta_{i,j}$, so the set of polynomials is orthonormal.

Question 4 Projection on Orthogonal Basis

Any quantity (function, vector, etc) can be represented as a linear combination of orthonormal basis components, with coefficients equal to the magnitude of a projection onto those bases. For functions on $x \in [-1, 1]$, using the Legendre polynomials as a basis, this expansion is

$$f(x) = \sum_l^{\infty} \frac{\langle f, P_l \rangle}{\langle P_l, P_l \rangle} P_l. \quad (24)$$

since $\langle P_l, P_l \rangle = 1$, the coefficients (“moments”) are $f_l = \langle f, P_l \rangle; l = 0, \dots, \infty$. Assuming that the series $\sum_{l=0}^L f_l P_l$ converges to $f(x)$ (that is, each successive term in the sequence of truncated reconstructions is closer to the true function than the previous term), the function can be approximated using only the first $L + 1$ terms.

4(a) The Legendre moments are

$$f_i = \langle f(x), P_i(x) \rangle = \int_{-1}^1. \quad (25)$$

Then, for $f(x) = 3x + 7$,

$$\begin{aligned} f_0 &= \int_{-1}^1 \frac{1}{\sqrt{2}} (3x + 7) dx \\ &= \frac{1}{\sqrt{2}} \left[\frac{3}{2}x^2 + 7x \right]_{-1}^1 \\ &= \frac{1}{\sqrt{2}} \left[\left(\frac{3}{2} + 7 \right) - \left(\frac{3}{2} - 7 \right) \right] \\ &= \frac{1}{\sqrt{2}} [14] \\ &= \frac{14}{\sqrt{2}} \\ f_0 &= 7\sqrt{2} \end{aligned} \quad (26)$$

$$\begin{aligned} f_1 &= \frac{\sqrt{3}}{\sqrt{2}} \int_{-1}^1 x (3x + 7) dx \\ &= \frac{\sqrt{3}}{\sqrt{2}} \left[x^3 + \frac{7}{2}x^2 \right]_{-1}^1 \\ &= \frac{\sqrt{3}}{\sqrt{2}} \left[\left(1 + \frac{7}{2} \right) - \left(-1 + \frac{7}{2} \right) \right] \\ &= \frac{2\sqrt{3}}{\sqrt{2}} \\ f_1 &= \sqrt{6} \end{aligned} \quad (27)$$

$$\begin{aligned}
f_2 &= \frac{\sqrt{5}}{\sqrt{2}} \int_{-1}^1 \frac{1}{2} (3x^2 - 1) (3x + 7) dx \\
&= \frac{1}{2} \frac{\sqrt{5}}{\sqrt{2}} \int_{-1}^1 (9x^3 - 3x + 21x^2 - 7) dx \\
&= \frac{1}{2} \frac{\sqrt{5}}{\sqrt{2}} \left[\frac{9}{4}x^2 - \frac{3}{2}x^2 + 7x^3 - 7x \right]_{-1}^1 \\
f_2 &= 0
\end{aligned} \tag{28}$$

4(b) For the function $f(x) = e^x$,

$$\begin{aligned}
f_0 &= \frac{1}{\sqrt{2}} \int_{-1}^1 e^x dx \\
&= \frac{1}{\sqrt{2}} [e^x]_{-1}^1 \\
f_0 &= \frac{1}{\sqrt{2}} [e^1 - e^{-1}] \\
f_1 &= \sqrt{\frac{3}{2}} \int_{-1}^1 x e^x dx \\
&= \sqrt{\frac{3}{2}} \left[x e^x \Big|_{-1}^1 - \int_{-1}^1 e^x dx \right] \\
&= \sqrt{\frac{3}{2}} [e^1 - (e^1 - e^{-1})] \\
f_1 &= \sqrt{\frac{3}{2}} e^{-1}
\end{aligned} \tag{30}$$

$$\begin{aligned}
f_2 &= \frac{1}{2} \sqrt{\frac{5}{2}} \int_{-1}^1 (3x^2 - 1) e^x dx \\
&= \frac{1}{2} \sqrt{\frac{5}{2}} \left[e^x (3x^2 - 1) \Big|_{-1}^1 - \int_{-1}^1 6xe^x dx \right] \\
&= \frac{1}{2} \sqrt{\frac{5}{2}} \left[[e^x (3x^2 - 1) - 6xe^x]_{-1}^1 + \int_{-1}^1 6e^x dx \right] \\
&= \frac{1}{2} \sqrt{\frac{5}{2}} [e^x (3x^2 - 6x - 1 + 6)]_{-1}^1 \\
&= \frac{1}{2} \sqrt{\frac{5}{2}} [e^x (3x^2 - 6x + 5)]_{-1}^1 \\
&= \frac{1}{2} \sqrt{\frac{5}{2}} [e^1 (3 - 6 + 5) - e^{-1} (3 + 6 + 5)] \\
f_2 &= \frac{1}{2} \sqrt{\frac{5}{2}} [2e^1 - 14e^{-1}]
\end{aligned} \tag{31}$$

4(c) For the function $f(x) = e^{5x}$,

$$\begin{aligned}
f_0 &= \frac{1}{\sqrt{2}} \int_{-1}^1 e^{5x} dx \\
&= \frac{1}{\sqrt{2}} \left[\frac{1}{5} e^{5x} \right]_{-1}^1 \\
f_0 &= \frac{1}{5\sqrt{2}} [e^5 - e^{-5}]
\end{aligned} \tag{32}$$